



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

Artificial Intelligence in the Contemporary World

[Project Domain: HR & Workload Management

Project Title: FacultyBalance-AI Assisted Faculty Time Allocation

Tools to be Used: Excel + Power BI + Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:



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Project Synopsis

A Collaborative Initiative — Equipping Faculty with Practical Skills in AI, Data Analytics, and Business Intelligence

Project Title	FacultyBalance: AI-Assisted Faculty Time Allocation
Domain	HR & Workload Management
Tools Used	Excel + Power BI + Microsoft Copilot

3. Problem Statement

A Higher Educational Institution depends on its efficiency on its core pillar, the teaching Faculty. Workload distribution of the faculty should ensure fairness, transparency, equity and align with institution’s goals. However, there are challenges in ensuring the workload balance which may impact reduced faculty productivity and thereby impacting the overall efficiency of the institutions.

Institutions lack a unified and data-driven mechanism to evaluate and balance faculty workload across teaching, research, and administrative responsibilities. Existing workload allocation practices are fragmented, manually managed, and often based on static timetables or subjective judgment. This leads to hidden inequities where some faculty members are overburdened while others are underutilized, directly impacting teaching quality, research productivity, and faculty satisfaction.

The absence of a centralized analytical system prevents institutions from identifying workload imbalances, benchmarking against expected norms, and making informed redistribution decisions. As a result, administrative decisions are reactive rather than optimized, increasing the risk of burnout, inefficiency, and suboptimal academic outcomes. There are web-based tools for equitable distribution of workload, but a lightweight model is the need of the hour. AI assisted workload summaries and imbalances are not yet incorporated into the existing tools.

This project addresses the need for a structured, data-driven decision support system that integrates faculty workload data into a unified framework, enabling real-time analysis, imbalance detection, and actionable insights through an interactive dashboard and present AI assisted summaries.

4. Objectives

1. To collect and understand the structure of the Faculty workloads dataset used in the project through Excel
2. To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values By Power query editor of Excel
3. To build interactive dashboards using Power BI
4. To leverage AI tools (Copilot) to automate or enhance the analysis and content generation process.
5. To develop a unified dataset capturing faculty teaching, research, and administrative responsibilities for accurate workload analysis.

6. To clean and preprocess the workload data, removing inconsistencies, standardising categories, and ensuring completeness
7. To design an interactive Power BI dashboard that visualizes workload distribution at individual & departmental levels and provides Teaching KPIS, Service Hours KPIS and Balance KPIS
8. To integrate AI and generate workload summaries at individual and departmental levels that can highlight imbalances if any and facilitate predictive modelling
9. To document the complete project process, upload all files to GitHub, and record a demonstration video for submission

5. Methodology / Implementation Plan

The following five-phase workflow provides a structured approach to completing this project.

#	Phase	Description
1	Data Collection & Exploration	Obtain and design a faculty workload dataset including fields such as Faculty name, Department, experience, number of subjects, Core/Elective subject, New/repetitive course, teaching hours, research activities, committee roles and hours required for events to be conducted & reported and administrative duties.
2	Data Cleaning & Preprocessing	Remove duplicates, fill missing values, standardise activity categories, and validate hour totals against the set norms using Excel and Power Query.
3	Analysis & AI Integration	Use Excel formulae to calculate total workload per faculty and by department. Apply Microsoft Copilot to generate workload summaries, identify outliers, and draft redistribution recommendations. Also enable predictive modelling for future workload planning
4	Visualisation / Output	Build a Power BI dashboard with workload distribution charts, faculty comparison views, department-level breakdowns, and KPI cards including Teaching KPIS, Research KPIS, Services KPIS, Balance KPIS. Document all Copilot prompts and outputs.
5	Documentation & Submission	Write the project report, upload all files to GitHub, and record a video demonstration. Submit all links for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop— Intel Core i3, 20 GB RAM
- Operating System: Windows 11 Pro
- Microsoft Excel : Excel 2019 or later
- Power BI Desktop — latest version (free download from Microsoft)
- Microsoft Copilot — via Microsoft 365 subscription
- Internet connection — 100mbps
- Web browser —Microsoft Edge (latest version)

B. Dataset

- Dataset needs to be sourced.
- Not from public sources.
- The dataset will be collected from our group of institutions and simulated for reasons of privacy.
- For AI prompt projects: description of the use case brief and reference content used as input

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	March 31, 2026
2	Data cleaning, preprocessing, and validation / Prompt design	April 3, 2026
3	Analysis, AI tool integration and insight generation	April 10, 2026
4	Dashboard / output build, testing, and review	April 15, 2026
5	Report writing, GitHub upload, and video recording and submission	April 20, 2026

7. Expected Outcomes / Conclusion

- A robust unified Dataset capturing hidden responsibilities and adoptable as a sector standard and publishable

- A low-cost, Excel–Power BI prototype for small and medium institutions to enable operational efficiency.
- An interactive workload dashboard combining subjective and objective data — a genuinely holistic management tool that no existing academic HR system currently provides.
- AI enabled Workload summaries and predictive modelling for workload planning to reduce imbalances and increase faculty productivity .

8. References

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